8 DETERMINE THE INITIAL SETTING TIME FOR THE GIVEN SAMPLE OF CEMENT

Aim

To find out the initial setting time of given sample of cement.

TJ87Y9

Apparatus and Materials Required

1. Vicat Apparatus
2. Needle (1 sq mm in c/s)
3. Mould
4. Non Porous Plate
5. Weighing Balance
6. Measuring Jar and Water
7. A Container
8. Trowel
9. Clock
10. Cement Sample (200 grams)

Procedure

1. Take 200 grams of cement
2. Fix the needle for initial setting time with the apparatus and adjust the pointer to point zero in the vicat's scale when the needle is placed above the plate.
3. Add 0.85 times of normal consistency value of water with the cement. Note the time (T_1) of adding water and prepare the paste.

Wt. of cement sample	= 200 grams
Percentage of water added	= 0.85 × Normal consistency value
	=
	=

4. Fill the prepared paste in the mould while it is placed above the plate.
5. Bring the needle just above the paste and allow it to penetrate the paste freely.
6. Note the measurement on the scale.
7. In the beginning, it penetrates completely. It is then taken out and dropped at a new place after some time.
8. This procedure is repeated until the pointer points 5mm to 7mm in the vicat's scale.
9. Note the time (T_2). Subtract the time noted in point 3 from it (T_2-T_1).
10. That is the time of initial setting for that sample.

XI_Civil_Practical_English.indd 237 12/21/2019 5:50:38 PM